SLE-3: Architectural Design (Full C4 Model)
Course: 02AML204 – Introduction to Artificial Intelligence
PRN: 25UAM049
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GitHub Link - https://github.com/kudalkargouri-crypto/SLE3.git

Project: Graph Search System Using BFS and DFS

1.System Title: Graph Search System Using BFS and DFS

The system performs graph search using Breadth-First Search (BFS) and Depth-First Search (DFS). The graph contains 800 nodes, with node 0 as the start node and node 799 as the goal node. BFS uses a queue, while DFS uses a stack. The system records execution time, the number of nodes expanded and whether the goal is found. This is the same project used for performance analysis in SLE-2.

2. Context Diagram (C4 Level 1)

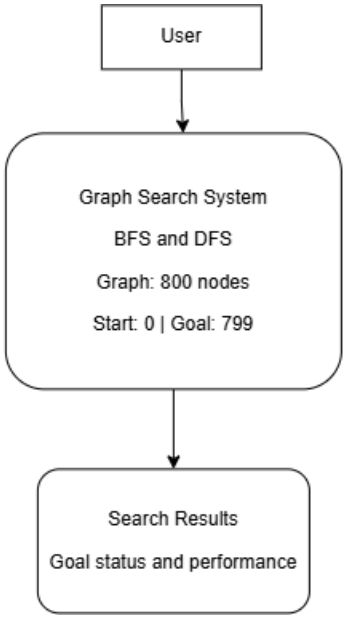


Figure 1: Context Diagram (C4 Level 1) of the BFS and DFS Graph Search System

3. Container Diagram (C4 Level 2)

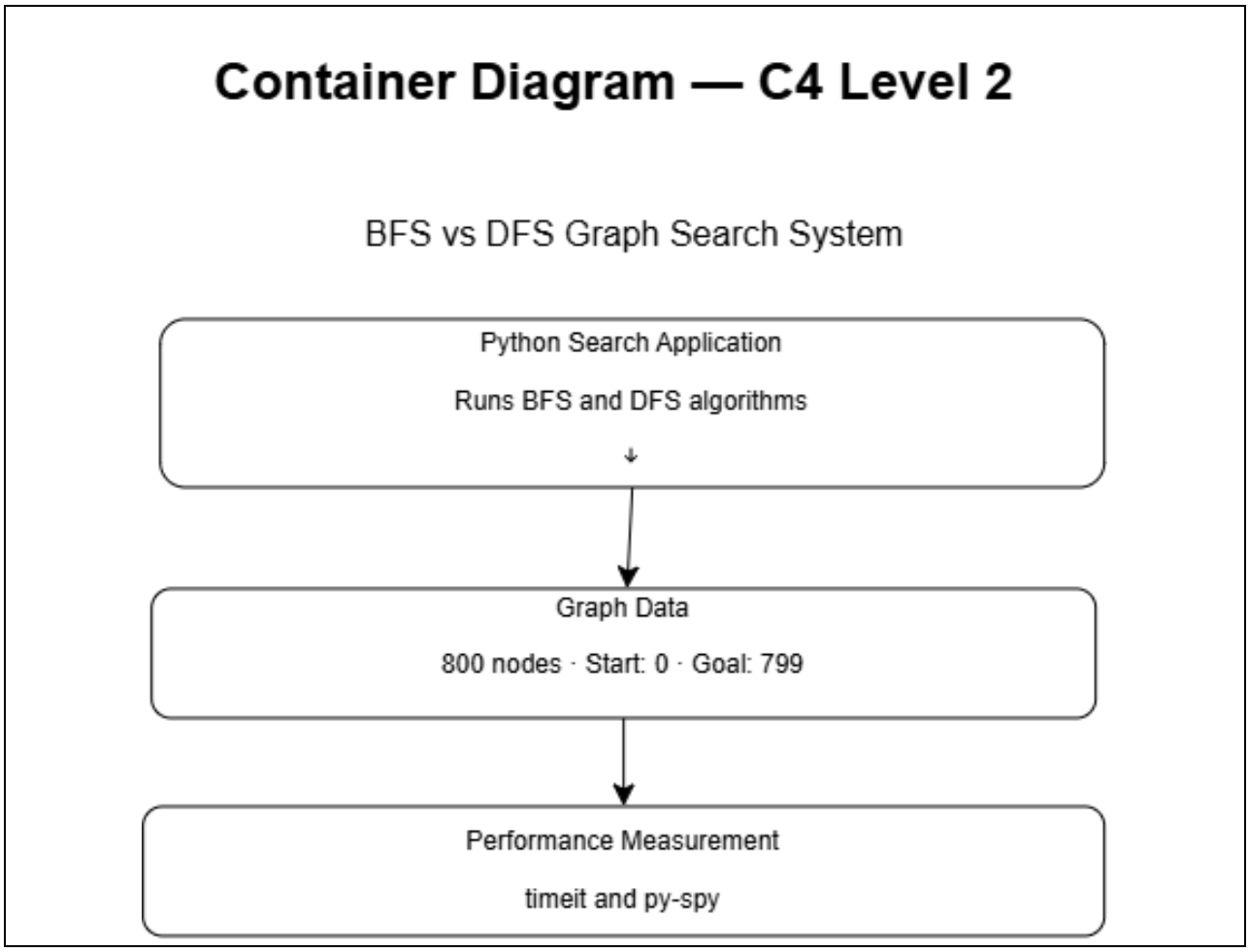


Figure 2: Container Diagram (C4 Level 2) of the BFS and DFS Graph Search System.

The Container Diagram shows the main parts of the BFS and DFS graph search system. The Python search application runs both algorithms on graph data containing 800 nodes. The program searches from node 0 to node 799. The execution time and program performance are measured using time it and py-spy.

Container	Responsibility
Input Module	Takes the start node and goal node/search input from the user.
Graph Management	Stores and manages the graph using the graph/adjacency-list representation.
Search Engine	Runs the selected BFS or DFS search algorithm.
Visited / Memory	Stores already visited nodes to avoid repeated exploration.

Output Module	Displays the final search result and search information.

4. Component Diagram (C4 Level 3)

The Component Diagram shows the internal components of the BFS and DFS Graph Search System. The BFS component uses a queue to explore nodes, while the DFS component uses a stack. The performance measurement component measures execution time and helps analyse the performance of both algorithms.

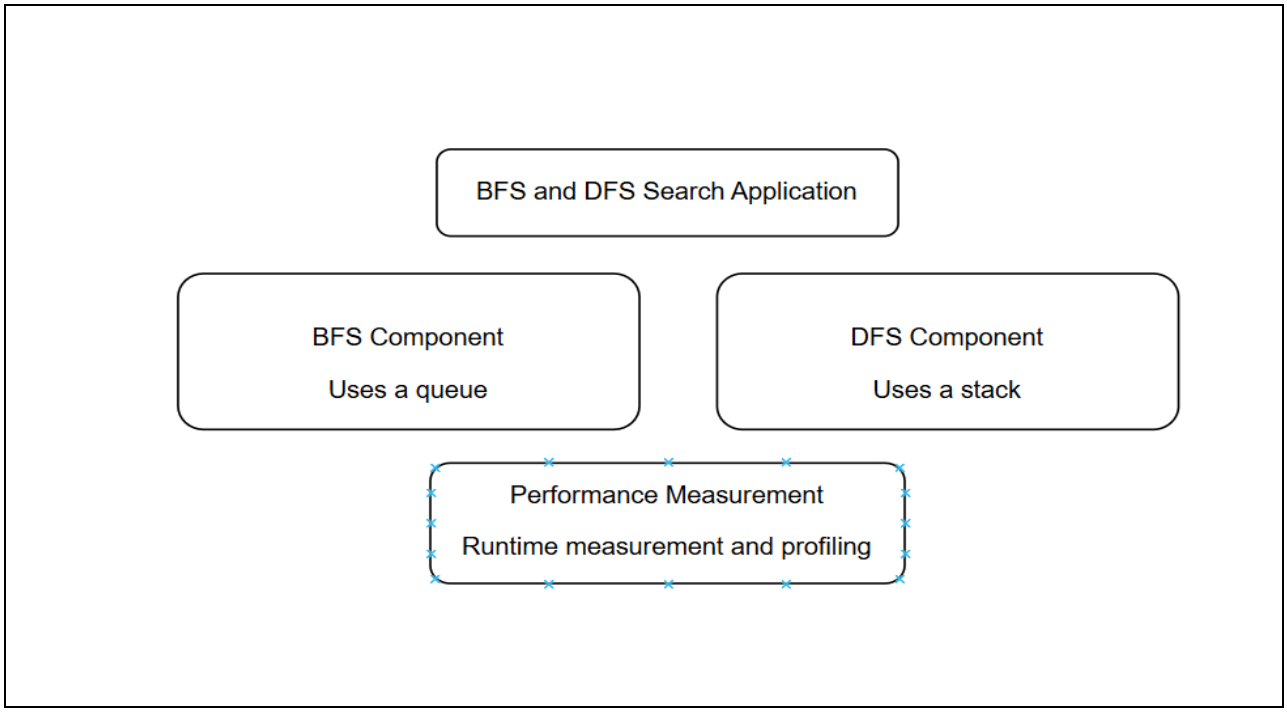


Figure 3: Component Diagram (C4 Level 3) of the BFS and DFS Graph Search System.

5. Code Level Overview (C4 Level 4):

The Code Level Overview describes the main functions and their responsibilities in the BFS and DFS Graph Search System. It explains how the graph is represented,

how BFS and DFS explore the nodes, how visited nodes are managed, and how the number of expanded nodes and execution time are measured.

File / Function	Responsibility
Graph representation	Stores the graph containing 800 nodes.
BFS function	Performs Breadth-First Search using a queue.
DFS function	Performs Depth-First Search using a stack.
Visited structure	Keeps track of visited nodes to avoid repeated exploration.
Node counter	Counts the nodes explored during the search.
Performance measurement	Measures execution time and supports performance analysis using `timeit` and `py-spy`.

6. Design Decisions

   The system separates graph representation from the BFS and DFS search algorithms so that both algorithms can be implemented and compared independently. BFS uses a queue to explore nodes level by level, while DFS uses a stack to explore deeper nodes first. A visited structure helps avoid repeated exploration of nodes. The system also measures execution time and uses py-spy for performance profiling. These design decisions support the performance analysis carried out in SLE-2.

7. Conclusion:
   This SLE helped me understand how a software system can be represented using the C4 architectural model. I learned about the four levels: Context, Container, Component, and Code Level. I also understood how BFS and DFS use different data structures to explore graph nodes. This architecture connects the implementation of the BFS and DFS algorithms with the performance analysis completed in SLE-2.